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of Science and Technology

Please find herewith:

- lists of additional literature for 1975 (list 9), 1976 (list 5) and 1977 (list 2)
- references to literature dealing with rearing of fleas
- a bibliography of publications on fleas (54 titles) by G. H. E. Hopkins (with some biographical notes)
- a concise report on the 1977 International Conference on Fleas

INTERNATIONAL CONFERENCE ON FLEAS - The enclosed concise report is merely a factual one, listing participants and events. It is worth mentioning here that a large room in the house was entirely devoted to a splendid display of the following items:

- colour photomicrographs illustrating histology of fleas (M. Rothschild)
- drawings of internal morphology, including nervous system (J. Schlein)
- electron photomicrographs of internal organs with Rickettsiae (S. Ito)
- photomicrographs of micro-organisms in fleas (H. E. Krampitz)
- photographs illustrating morphology of Ceratophyllid fleas (R. Traub)
- stereoscan photomicrographs of external structures (M. T. Greenwood)
- various stereoscan photomicrographs (F. G. A. M. Smit)
- distribution and host data of New Zealand fleas (R. L. C. Pilgrim)
- distribution maps of British fleas (R. S. George)
- a flea-jump counter (S. Sternberg)
- new microscopes and equipment (Carl Zeiss)
- two historical fleas from the Rothschild collection (F. G. A. M. Smit)

For the first conference on fleas, two years ago in Lund, Sweden, most of the work was shouldered by two ladies - Mrs Gunvor Brinck-Lindroth and Miss Gunilla Lindqvist. This time, too, two ladies have done all the hard work, namely Dr Miriam Rothschild and Miss José Beer. After the final lecture on Saturday June 25th, Professor Brinck - on behalf of all delegates - was therefore pleased to hand Dr Rothschild, the generous hostess of the Conference, a memento in the form of an engraved goblet while Mr Smit presented Miss Beer with a framed old map of Northamptonshire (on which Ashton) surrounded by signatures of the delegates.

Those who were fortunate enough to participate must have memories of a fruitful and delightful week - a unique experience not easily forgotten.

One colleague summed the meeting up very succinctly when she wrote:
"The Conference was great, just great."

Here are two personal observations (not meant to be criticisms) of a general nature - they would apply to any lecture-meeting:

- Good (in both quality and quantity) projected illustrations illuminate the dulllest or most brilliant text, especially before an international audience. However, it seems rather useless to project complicated tables or graphs (for which there appears to be a certain fondness) as these can seldom be digested, or even read, during the short time that they are on the screen.
- In any lecture hall there should be notices "Please try not to smoke". Smokers (I have been one myself) do not realize how extremely unpleasant it can be for non-smokers to have to breathe in the foul air which addicts rather selfishly polluted.

F.S.

CORRECTION - Flea Conference Report, reverse page, line 4 from above:
HOOGSTRAAT should be HOOGSTRAAL
* * *

REPRINTS are AVAILABLE of the papers listed below. You are welcome to ask for any of these which you may need: -

- of the JORDAN publications, as listed in the bibliography sent with FN 11:
1921a 1921b 1929b 1931b 1932j 1936f 1938a 1938b 1938d 1938e 1939a
1939e 1939f 1940b 1941a 1941b 1941c 1942b 1942c 1943a 1944 1945a
1945b 1946a 1946b 1947a 1947b 1951a 1951b 1954 1958

- of the HOPKINS publications, as listed in the enclosed bibliography:
1948b 1951c 1952c 1952d 1952e 1952f 1954a 1955 1956 1958c 1958e
1959 1961a 1961b 1962a 1962b 1962c 1963 1964a 1964b 1965a 1965b
Hopkins & Johnson 1959 Hopkins & de Meillon 1964

All requests for reprints will be kept till about the end of this year in order to give everyone a fair chance to apply. In case supplies of certain items prove to be insufficient to meet demand, the available copies will be allocated by luck of draw.
* * *

REARING - Working with live fleas - for many a purpose - can be most rewarding as well as frustrating. Over the years I have noticed an increase in the frequency of requests for references to literature in which techniques of rearing fleas are dealt with. It seems opportune, therefore, to append a list of the important articles on the subject; this is not claimed to be a complete list - if you spot glaring omissions, please let me know.

The literature listed (in chronological order) on a separate sheet contains descriptions (some of which not very detailed) of methods used to rear (usually "in the laboratory") the following species of fleas [the numbers following the names of the fleas are those preceding the titles in the list of references]:

Amalaraeus penicilliger 28	Nosopsyllus consimilis 34 35
Amphipsylla primaris mitis 28	Nosopsyllus fasciatus 3 11 20 22 27 31
Ceratophyllus anisus 28	34 35
Ceratophyllus gallinae 34	Nosopsyllus laeviceps 31 34
Ceratophyllus sciurorum 27 34	Nosopsyllus tersus 34
Citellophilus tesquorum 28 31	Opisodasys nesiotus 20
Coptopsylla sp. 34	Oropsylla idahoensis 20
Ctenocephalides canis 2 6 15 18	Oropsylla rupestris 38
Ctenocephalides felis felis 6 15 17	Oropsylla silantiewi 28
21 22 25 27 34	Orchopeas howardi 9
Ctenophthalmus assimilis 28	Orchopeas s. sexdentatus 20
Ctenophthalmus dolichus 34	Polygenis bohlsi jordani 13
Ctenophthalmus orientalis 35	Pulex irritans 1 2 3 26 34 39
Ctenophthalmus wagneri 42	Pulex simulans 26
Ctenophthalmus wladimiri 43 44	Spilopsyllus cuniculi 23 41
Diamanus montanus 1 20	Stenoponia tripectinata insperata 22
Echidnophaga gallinacea 4 30	Tunga penetrans 10
Frontopsylla elata caucasica 40	Xenopsylla astia 5 11 22
Frontopsylla luculenta 28	Xenopsylla brasiliensis 16
Leptopsylla pavlovskyi 28	Xenopsylla cheopis 3 5 11 12 13 14 16
Leptopsylla segnis 22 28 32 35	19 20 21 22 24 28 29 30 31 33 35
Malaraeus telchinus 20	36 37
Megabothris abantis 20	Xenopsylla conformis 22 31
Megabothris turbidus 27 34	Xenopsylla hirtipes 34
Neopsylla bidentatiformis 28	Xenopsylla nuttalli 34
Neopsylla pleskei 28	Xenopsylla skrjabini 34
Neopsylla setosa 31 34	Xenopsylla vexabilis 12 24

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REFERENCES to descriptions of METHODS OF REARING FLEAS

1. Mitzmain, M.B., 1910 - General observations on the bionomics of the rodent and human fleas. Bull. publ. Hlth Mar.-Hosp. Serv. (38): 1-34.
2. Harms, B., 1912 - Untersuchungen über die Larve von *Ctenocephalus canis* Curtis. I. Teil. Arch. mikrosk. Anat. EntwMech. 80(1): 167-216, figs. 1-13, pl. XIII figs. 1-16. [rearing: 172-175].
3. Bacot, A.W., 1914 - Reports on plague investigations in India. LXIX. A study of the bionomics of the common rat fleas and other species associated with human habitations, with special reference to the influence of temperature and humidity at various periods of the life history of the insect. J. Hyg., Camb. 13 (Suppl. 3): 447-654, figs. 1-3, pls. XXVII-XXXIV. [rearing: 449-470].
4. Parman, D.C., 1923 - Biological notes on the hen flea, *Echidnophaga gallinacea*. J. agric. Res. 23(12): 1007-1009.
5. Hirst, L.F., 1926 - Researches on the parasitology of plague. Part 1. Ceylon J. Sci. (D) 1(4): 155-271, figs. 1-2, pl. XXVI figs. 1-6. [rearing: 209-213].
6. Rifenburgh, S.A., 1927 - Raising fleas for laboratory purposes. Proc. Indiana Acad. Sci. 36: 311-312.
7. Hase, A., 1929 - Methoden der Züchtung von Wanzen, Läusen und Flöhen. In: W. Kollé & A. von Wassermann - Handb. pathogenen Mikro-organismen. (ed. 3). 10(37): 591-616, figs. 1-14. [Siph.: 610-614].
8. Pavlovsky, Ye.N., 1931 - Sammeln, Züchtung und Untersuchung der Flöhe. In: E. Abderhalden - Handb. biol. Arbeitsmethoden. Abt. IX, Tl 7, Heft 1, Liefg 346: 97-160, figs. 1-96. [rearing: 107-113].
9. Sikes, E.K., 1931 - Notes on breeding fleas, with reference to humidity and feeding. Parasitology 23(2): 243-249.
10. Hicks, E., 1932 - A simple apparatus for breeding fleas. Ann. trop. Med. Parasit. 26(2): 147-148, figs. 1-2.
11. Leeson, H.S., 1932 - Methods of rearing and maintaining large stocks of fleas and mosquitos for experimental purposes. Bull. ent. Res. 23(1): 25-31, figs. 1-5.
12. Eskey, C.R., 1934 - Epidemiological study of plague in the Hawaiian Islands. Bull. U.S. publ. Hlth (213): 1-70. [rearing: 28-29, 35, 39-40].
13. Hecht, O., 1943 - Algunas consejos tecnicos para la cria de pulgas. Rev/ta Sanid. Asist. soc., Caracas 8(6): 1163-1165.
14. Hollenbeck, A.H., 1946 - A practical method for mass production and transfer of *Xenopsylla cheopis*. J. Parasit. 32(5): 463-464, fig. 1.
15. Linduska, J.P. & Cochran, J.H., 1946 - A laboratory method of flea culture. J. econ. Ent. 39: 544-545.
16. Edney, E.B., 1947 - Laboratory studies on the bionomics of the rat fleas, *Xenopsylla brasiliensis* Baker and *X. cheopis* Roths. III. Further factors affecting adult longevity. Bull. ent. Res. 38(3): 389-404.
17. Bruce, W.N., 1948 - Studies on the biological requirements of the cat flea. Ann. ent. Soc. Amer. 41(3): 346-352.
18. Woke, P.A., 1951 - A rabbit-ear cage for bloodsucking Arthropods. Publ. Hlth Rep. Wash. 66(15): 464-471, figs. 1-2.
19. Kilpatrick, J.W. & Fay, R.W., 1952 - DDT-resistance studies with the oriental rat flea. J. econ. Ent. 45(2): 284-288, figs. 1-3.
20. Burroughs, A.L. & Britschgi, M.B., 1953 - Sylvatic plague studies. X. Survival of rodent fleas in the laboratory. Parasitology 43(1-2): 35-48.
21. Smith, C.N. & Eddy, G.W., 1954 - Techniques for rearing and handling body lice, oriental rat fleas, and cat fleas. Bull. Wld Hlth Org. 10(1): 127-137, figs. 1-6.
22. Baltazard, M. & Eftekhari, M., 1957 - Techniques de récolte, de manipulation et d'élevage des puces de rongeurs. Bull. Wld Hlth Org. 16(2): 436-440.
23. Rothschild, M., 1957 - Breeding the rabbit flea (*Spilopsyllus cuniculi* (Dale)) in captivity. Entomologist 90(1135): 304-305.
24. Stark, H.E. & Kartman, L., 1957 - Studies on *Pasteurella pestis* in fleas. VI. The laboratory culture of *Xenopsylla vexabilis hawaiiensis* Jordan, 1932. Amer. J. trop. Med. Hyg. 6(4): 707-711.

25. Hudson, B.W. & Prince, F.M., 1958 - A method for large-scale rearing of the cat flea, *Ctenocephalides felis felis* (Bouché). Bull. Wld Hlth Org. 19: 1126-1129, figs. 1-3.
26. Hudson, B.W. & Prince, F.M., 1958 - Culture methods for the fleas *Pulex irritans* (L.) and *Pulex simulans* Baker. Bull. Wld Hlth Org. 19: 1129-1133, figs. A-C.
27. Kir'yakova, A.N., 1961 - Laboratory methods for the study of the reproductive biology of fleas. Ent. Obozr., Moskva 40(2): 443-447. [translation: 1961, Ent. Rev. (Wash.) 40(2): 232-233].
28. Vasil'yev, G.I., 1961 - Observation on rearing fleas in laboratory conditions. Dokl. Irkutsk. protivochumn. Inst. 2: 97-99.
29. Sternberg, S. & Tahori, A.S., 1962 - Control of *Aleuroglyphus ovatus* (Troupeau) in flea cultures. Bull. ent. Res. 53(2): 257-264.
30. Suter, P.R., 1964 - Biologie von *Echidnophaga gallinacea* (Westw.) und Vergleich mit andern Verhaltenstypen bei Flöhen. Acta trop. 21(3): 193-238, figs. 1-15. [rearing: 196-199].
31. Kolpakova, S.A., Shatas, Ya.F., Lapina, N.F. & Luk'yanova, A.D., 1965 - From the practice of laboratory maintenance of rodent fleas for experimental investigations. Epidemiol. Epizootol. osobo opasn. Infektsii, Moskva: 287-299.
32. Krampitz, H.E., 1965 - Beobachtungen an einer Laboratoriumszucht von *Leptopsylla segnis* Schönherr, 1811. Z. Parasitenk. 26: 197-214, figs. 1-3.
33. Krishnamurthy, B.S., 1966 - Rat fleas. In: C.N. Smith - Insect colonization and mass production. New York & London (Academic Press): 73-83, figs. 1-7.
34. Vysotskaya, S.O. & Kir'yakova, A.N., 1970 - Methods of collecting and studying fleas and their larvae. Leningrad (Izd. "Nauka"): 83 pp., 26 figs. [rearing: 33-49].
35. Yurgenson, I.A. & Il'ina, S.V., 1970 - Application of pelleted concentrated food for standard cultivation of fleas. Parazitologiya 4(1): 70-71.
36. Cavanaugh, D.C., Stark, H.E., Marshall, J.D. & Rust, J.H., 1972 - A simple method for rearing fleas for insecticide testing in the field. J. med. Ent. 9(1): 113-114.
37. Cole, M.M., VanNatta, D.L., Ellerbe, W. & Washington, F., 1972 - Rearing the oriental rat flea. J. econ. Ent. 65(5): 1495-1496, figs. 1-3.
38. Hilton, D.F.J., 1972 - A method for rearing fleas of ground squirrels. Trans. R. Soc. trop. Med. Hyg. 66(1): 188-189.
39. Zolotova, S.I. & Yakunin, B.M., 1973 - The development of *Pulex irritans* L., 1758 under experimental conditions. Parazitologiya 7(1): 24-30.
40. Guseva, A.A. & Kosminsky, R.B., 1974 - Feeding and reproduction of *Frontopsylla elata caucasica* Ioff et Arg., 1934 under experimental conditions. Osobo opasn. Infektsii Kavkaza (Stavropol') 1: 132-134.
41. Sobey, W.R., Menzies, W. & Conolly, D., 1974 - Myxomatosis: some observations on breeding the European rabbit flea *Spilopsyllus cuniculi* (Dale) in an animal house. J. Hyg., Camb. 72(3): 453-465, figs. 1-6.
42. Kosminsky, R.B. & Guseva, A.A., 1975 - Feeding and reproduction of *Ctenophthalmus wagneri* Tifl., 1928 in experiment. Parazitologiya 9(3): 265-270.
43. Talybov, A.N., 1975 - The life duration of *Ctenophthalmus wladimiri* Is.-Gurv., 1948 under laboratory conditions. Parazitologiya 9(4): 354-358.
44. Talybov, A.N., 1976 - The development of the immature stages of the flea *Ctenophthalmus wladimiri* Is.-Gurv., 1948. Parazitologiya 10(4): 320-324.

(It will be understood that -although their titles are given in English above- papers in Russian periodicals are in Russian.)

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(compiled by F.S.
VII.1977)

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1975 (list 9)

- Dipeolu, O.O. - Ectoparasites of local pigs in western Nigeria. Nigerian J. anim. Production 2(2): 222-226.
- Dyatlova, T.I. & Korneyeva, L.A. - The influence of culture conditions on changes of a morpho-physiological character in the eastern rat flea (*Xenopsylla cheopis*). Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 222.
- Gafarova, F.A. & Gadzhiyev, A.T. - Ectoparasites of *Mus musculus* in Azerbaydzhan. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 210-211.
- Khodykina, Z.S. - The fat body of the rat fleas *Xenopsylla cheopis* and *Ceratophyllus fasciatus*. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 263-264.
- Labunets, N.F., Gubareva, N.P., Syrvacheva, N.G., Dyevkin, A.I. & Kazakova, T.I. - Fleas (Siphonaptera) of rodents on the northern slopes of central Caucasus. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 235-236.
- Nazarova, I.V. & Yarullova, R.A. - On the geographical variation of the flea *Ceratophyllus penicilliger*. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 244.
- Nyel'zina, Ye.N., Myedvedev, S.I., Pichka, V.E., Protopopyan, M.G., Bakhtinova, N.Z., Zakrutkin, A.Ye. & Kuznetsov, M.V. - Seasonal aspects in the structure of burrow microbiocenoses of the large gerbil (*Rhombomys opimus*) in the Uralo-Embinsk inter-river area. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 245.
- Osipova, N.Z. - Fleas and other ectoparasites of gerbils in western Tyan'-Shan'. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 245-246.
- Vashchyonok, V.S. - The activity of the digestion of blood in fleas. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 209.
- Veen, T.W.S. van & Mohammed, A.N. - Louse and flea infestations on small ruminants in the Zaria area. J. Nigerian vet. med. Ass. 4(2): 93-96, figs. 1-2.
- Zhovtyi, I.F. - Population ecology of fleas (Siphonaptera) of Siberia and the Far East - results and problems. Mater. 7 S'yezda vseyesoyuzn. ent. Obshch. (1): 224-225.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1976 (list 5)

- Beaucournu, J.C. & Gosalbez, J. - Contribution à l'étude des Siphonaptères de Catalogne française et Espagnole. Vie & Milieu (c) 25(1-C): 69-86, maps 1-2.
- Bonin, W. - Heptenophos - a new insecticide for the control of ectoparasites. Proc. 8th Brit. Insecticide Fungicide Conf. (Brighton) 2: 705-709.
- Iqbal, Q.J. & Humphries, D.A. - Remating in the rat flea *Nosopsyllus fasciatus* (Bosc). Pakistan J. Zool. 8(1): 39-41.
- Isaacs, M., Qhobela, Q.M., Davis, D.H.S. & Growngold, T.P. - Serological studies on human plague in southern Africa. Part II. A bubonic/pneumonic plague epidemic in Lesotho. S. Afr. med. J. 50(24): 929-932, figs. 1-2.
- Jurík, M. - *Monopsyllus sciurorum* (Schrank 1781) ako parazit v chovoch noriek v CSSR. Zprávy Čsl. zool. Spol. 7-9: 11-14.
- Kozlov, M.P. & Rozanova, G.N. - The dependence of the formation of plague blocks in fleas on specific particularities of erythrocytes of animals. Probl. osobo opasn. Infektsii (Saratov) 5(51): 53-57.
- Pérez-Laclette, T. & Barrera, A. - Consideraciones sobre la presencia de *Pleochaetis dolens* (Jordan y Rothschild) 1914, en México (Siphonaptera Ceratophyllidae). Folia ent. mex. (36): 99-100.
- Wegner, Z. & Kruminis-Łozowska, W. - Rodents in warehouses of the port of Gdynia and their ectoparasites investigated in the year 1974-1975. Bull. Inst. marit. trop. Med. Gdynia 27(3-4): 321-329.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1977 (list 2)

- Anonymous - Statens Skadedyrlaboratorium. Årsberetning 1975. Lyngby. 80 pp., figs. (Siph.: 16).
- Anonymous - Control of fleas. Publ. Canada Dept. Agric. (1187): ? pp., 1 fig.
- Bang, P. & Hallas, T. - Lopper på danske smågnavere. (Fleas on Danish small mammals). Statens Skadedyrlaboratorium. Årsberetning 1975. Lyngby: 50-51.
- Beaucournu, J.C. & Rosin, G. - *Hystrihopsylla talpae* (Curtis, 1826) (Siphonaptera, Hystrihopsyllidae), relique pliocène au Maroc. Annls Parasit. hum. comp. 52(1): 101-103.
- Byerendyayeva, E.L., Shvarts, Ye.A. & Semenova, N.I. - A contribution to the fauna of gerbil fleas in the eastern Ferganskaya oblast'. Ekol. med. Znach. Peshchanok Fauny SSR (Moskva): 249-250.
- Cerqueira, E.J.L. & Linardi, P.M. - Indices pulicidians em Belo Horizonte, Minas Gerais, Brasil. Ciênc. Cult., S. Paulo 29(2): 191-193.
- Garbuzov, V.K. - On the ecology of the giant mole rat (*Spalax giganteus* Nehr.) in Kazakhstan. Byull. mosk. Obshch. Ispyt. Prir., Biol. 82(2): 32-37.
- Haarlov, N. - Ectoparasites (Mallophaga, Siphonaptera, Acarina) from birds of Jan Mayen Island, Norway. Norway J. Ent. 24(1): 37-41.
- Kunitskaya, N.T., Kunitsky, V.N., Gauzshtein, D.M. & Savelova, N.M. - The physiological age of fleas and an analysis of the age composition of a natural population of *Xenopsylla gerbilli* Wagn. Parazitologiya 11(3): 202-209. figs. 1-4.
- Kunitsky, V.N., Yakunin, B.M., Ilyushkina, V.I., Karinsky, A.N., Lenikova, Z.F., Myedbedevskikh, V.I. & Chyornova, Z.F. - The density of the flea populations of the large gerbil in eastern Caspian lowlands, west Ustyurt and the Mangyshlak peninsula. Ekol. med. Znach. Peshchanok Fauny SSR (Moskva): 266-268.
- Mitchell, J.A. - Studies on the biology of *Trypanosoma* (H.) *peromysci* Watson, 1912 (Protozoa: Kinetoplastida: Trypanosomidae). Diss. Abstr. (B) 37(7): 3299B.
- Ryba, J. & Balát, F. - Fleas infesting nests of *Passer montanus* and some other birds. Folia parasit., Praha 24(2): 191-192.
- Shepherd, R.C.H., Nolan, I.F., Lane, I.L. & Edmonds, J.W. - Ectoparasites of the European hare (*Lepus europaeus* (Pallas)) in two regions of Victoria. J. aust. ent. Soc. 16(1): 1-5.
- Sokolova, A.A. - Seasonal dynamics in the numerosity of *Xenopsylla gerbilli* in Muyunkumy. Ekol. med. Znach. Peshchanok Fauny SSR (Moskva): 282-283.
- Syedin, B.V. - On plague infectivity of fleas in colonies of the large gerbil in the Karakumy lowlands. Ekol. med. Znach. Peshchanok Fauny SSR (Moskva): 347-349.
- Thompson, G.B. - Ectoparasites from mammals. J. Oman Studies, Special Report (Scient. Results Oman Flora Fauna Survey 1975): 227-228.
- Thompson, M.J.A. - The pipistrelle bat (*Pipistrellus pipistrellus*) in and around York. Naturalist 102(942): 109-116. (Siph.: 115-116).
- Vysotskaya, S.O. - The microbiocenosis of nests of the Transcarpathian squirrel (*Sciurus vulgaris carpathicus* P.). Parazitol. Sb. 27: 181-191.
- Yakuba, V.N., Lazareva, L.A., Klimov, V.T., Mayevsky, M.P., Bondarenko, A.A. & Mashkovsky, I.K. - The flea *Ceratophyllus fasciatus* as a vector of the Gorno-Altai strain of plague microbe. Parazitologiya 11(3): 268-270.

List of the publications on SIPHONAPTERA by G. H. E. HOPKINS

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AM - Ann. Mag. nat. Hist.	MM - Entomologist's mon. Mag.
BZ - Bull. zool. Nomencl.	PA - Parasitology, Camb.
EA - E. afr. med. J.	PE - Proc. R. ent. Soc. Lond.
EN - Entomologist	PW - Proc. ent. Soc. Wash.
JH - J. Hyg., Camb.	RM - Rec. med. Res. Lab. Kenya
JW - J. Wash. Acad. Sci.	TE - Trans. R. ent. Soc. Lond.
ML - Mem. Lond. Sch. trop. Med. Hyg.	UJ - Uganda J.

- 1935 (9.XI) - Some observations on the bionomics of fleas in East Africa.
PA 27(4): 480-488.
- 1938 (30.III) - Cotton and plague in Uganda. JH 38(2): 233-247.
- 1941 (IV) - Methods of control of plague in Uganda. EA 18: 18-23.
- 1947a (IX) - Annotated and illustrated keys to the known fleas of East Africa.
UJ 11(2): 133-190, pls I - XXI (figs. 1-146).
- 1947b (31.XII) - *Dinopsyllus longifrons* and *D. lypusus* (Siphonaptera).
EN 80(1015): 284-286.
- 1948a (13.VII) - The genotype of *Trichopsylla Kolenati*, 1863. EN 81(1022):
168-169.
- 1948b (12.XI) - The identity of *Pulex metallescens Kolenati* 1856.
EN 81(1026): 249-250.
- 1949a (26.VI) - Notes on fleas. EN 82(1033): 134-138.
- 1949b (21.XI) - A new South African *Xenopsylla* (Siphonaptera, Pulicidae).
EN 82(1038): 251-253, figs. 1-4.
- 1949c - Report on rats, fleas and plague in Uganda. (Government Printer,
Uganda). 52 pp., 23 tables, 1 map.
- 1950a (16.II) - The type species of *Trichopsylla Kolenati*: a further note.
EN 83(1041): 31-33.
- 1950b (13.VII) - The identity of two species described by *Kolenati* in 1859.
EN 83(1046): 162-163.
- 1950c (8.IX) - An awkward problem in flea nomenclature. EN 83(1048): 201-202.
- 1951a (19.IV) - An awkward problem in flea nomenclature: a supplementary note.
EN 84(1055): 90-91.
- 1951b (VI) - Notes on fleas. AM (12)4 (42): 529-544, figs. 1-5.
- 1951c (15.VIII) - Proposed use of the plenary powers to validate the specific
trivial name "*acuminata*" Ioff & Tiflov, 1946 (as published in the combi-
nation "*Rhadinopsylla* (*Rectofrontia*) *acuminata*") as applied to the spe-
cies numbered "68" by those authors (Class Insecta, Order Siphonaptera).
BZ 2(9-10): 296-297.
- 1951d (4.IX) - The scientific name for the fleas. EN 84(1060): 208-214.
- 1952a (7.II) - Two overlooked descriptions of Japanese fleas. EN 85(1065): 33.
- 1952b (14.V) - The type and type-locality of *Neopsylla bidentatiformis*
(Wagner), 1893 (Siphonaptera). EN 85(1068): 102-104.
- 1952c (23.VII) - Proposed use of the plenary powers to designate a type species
for the genus "*Palaeopsylla*" Wagner, 1903 (Class Insecta, Order Siphon-
aptera) in harmony with the generally accepted use of that name. BZ 6(8):
239-241.

- 1952d (29.VIII) - Proposed use of the plenary powers to suppress the generic name "*Trichopsylla*" Kolenati, 1863 (Class Insecta, Order Siphonaptera) for the purposes of Article 25 but not for those of Article 34. BZ 6(11): 349-352.
- 1952e (29.VIII) - Proposed use of the plenary powers to suppress two specific trivial names in the Order Siphonaptera (Class Insecta). BZ 6(11): 353-354.
- 1952f (17.XI) - Notes on synonymy in Siphonaptera. JW 42(11): 363-365.
- 1954a (28.X) - Nomenclatorial notes on the flea of the grey squirrel. EN 87(1096): 197-198.
- 1954b (15.XI) - The male and systematic position of *Epitedia testor* (Rothschild). EN 87(1097): 217-219, figs. 1-2.
- 1955 (6.XII) - Bibliography of scientific publications by H. E. K. Jordan. TE 107: 77-94.
- 1956 (9.V) - Proposed use of the plenary powers to secure the interpretation of the nominal species "*Palaeopsylla daea*", Dampf, 1910 (Class Insecta, Order Siphonaptera) in a manner consistent with the universally accepted use of that name. BZ 11(11): 349-352.
- 1957 (18.IV) - Host-associations of Siphonaptera. Neuchâtel (1st Symp. Host Specif. Parasites Vertebr.): 64-87.
- 1958a (8.I) - Submission of a draft text of a provision relating to the method to be followed in citing the date and authorship to be attributed to a substitute family-group name. BZ 15(7-8): 185-186.
- 1958b (8.I) - Submission of a draft text providing for the reversal of the provisions in Copenhagen Decision 54(1)(b) in relation to the date and authorship to be attributed to a substitute family-group name. BZ 15(7-8): 255-256.
- 1958c (21.III) - Order/Class group names in zoology with special reference to the name to be used for the Order of Insects comprising the fleas. BZ 15(16-17): 547-552.
- 1958d (27.VI) - Order/Class group names for the fleas. BZ 15(31-33): 1069.
- 1958e (28.XI) - Order-group and family-group names for the fleas. AM (13)1(7): 479-487.
- 1959 (22.IX) - Family-group names for Siphonaptera: some corrections. AM (13)1(12): 801.
- 1961a (8.V) - More family-group names in Siphonaptera. AM (13)3(35): 643-644.
- 1961b (2.VI) - Insects of Micronesia. Siphonaptera. Ins. Micronesia 14(4): 91-107, figs. 1-9.
- 1962a (16.II) - The provenance of certain fleas reputedly from Myitkyina. EN 95(1185): 54-57.
- 1962b (18.V) - [Review of] B. de Meillon, [D.] H. S. Davis and F. Hardy, 1961, Plague in southern Africa. Volume I: The Siphonaptera (excluding Ischnopsyllidae). EN 95(1188): 134-136.
- 1962c (23.XI) - Type-material of some fleas described by Weyenbergh. EN 95(1194): 265-267.
- 1963 (26.IV) - *Ctenophthalmus* Kolenati, 1856 (Insecta, Siphonaptera). Proposed retention under the plenary powers and related matters. BZ 20(3): 217-223, pl. 4.

- 1964a (17.VIII) - Selection of a lectotype for *Sarcopsyllus gallinaceus* Westwood, 1875 (Siphonaptera). MM 99(1194-1195): 215.
- 1964b (22.X) - Notes on fleas described by Curtis in "British Entomology". EN 97(1217): 227-231.
- 1965a (2.XI) - Application for the stabilization of the name *Ceratophyllus soricis* Dale, 1878 (Insecta, Siphonaptera) by establishment of a neotype. BZ 22(4): 233-234.
- 1965b (2.XI) - The type-species of *Monopsyllus Kolenati*, 1857 (Insecta, Siphonaptera) and related matters. BZ 22(4): 250-254.

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- HOPKINS, G. H. E. & MEILLON, B. DE, 1964 (30.IV) - Two species of Epirimiinae (Siphonaptera: Chimaeropsyllidae) from South West Africa. PE (B)33(3-4): 39-43, figs. 1-5.
- HOPKINS, G. H. E. & ROTHSCCHILD, M., 1953 (V) - An illustrated catalogue of the Rothschild collection of fleas (Siphonaptera) in the British Museum (Natural History). London (The Trustees of the British Museum). I. Tungidae and Pulicidae. xv + 361 pp., frontispiece, 466 figs., 45 pls.
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- HOPKINS, G. H. E. & ROTHSCCHILD, M., 1966 (24.XII) - An illustrated catalogue of the Rothschild collection of fleas (Siphonaptera) in the British Museum (Natural History). London (Trustees of the British Museum (Natural History)). IV. Hystrichopsyllidae (Ctenophthalminae, Dinopsyllinae, Doratopsyllinae and Listropsyllinae). viii + 549 pp., 926 figs, 12 pls.
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- HOPKINS, G. H. E. & TRAUB, R., 1955 (6.XII) - The genus *Cratynius* Jordan (Siphonaptera) and its systematic position, with a description of a new species. TE 107: 249-264, figs. 1-17.
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- SYMES, C. B. & HOPKINS, G. H. E., 1932 (I) - Notes on fleas of rats and other hosts in Kenya. RM (1): 1-57.

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NOTE

H. 1949c - The following anecdote - which much amused Hopkins at the time - relating to this paper appeared in The Uganda Herald in January, 1952: "In the excellent new stationery store [of the Government Press at Entebbe], built last year, are shelved millions of forms and many Government reports, stacked on rat-proof steel shelving. Despite this, there was clear proof that a hungry rat had made a meal of some of H. M. Stationery. The main damage was done to a bundle of pamphlets by Mr G. H. E. Hopkins, entitled "Rats, fleas and plague in Uganda."

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Biographical notes on GEORGE HENRY EVANS HOPKINS

Born: 22.III.1898 at Hanley, Staffordshire, England.

Educated at Rossall, 1911-1916; Downing College, Cambridge, 1920-1923. B.A. (2nd Class Honours) in Natural Science Tripos in 1923, M.A. in 1926 (took elementary courses in agriculture (including tropical) while at Cambridge, and entomology under F. Balfour-Browne).

Served as Lieutenant in the 4th Battalion South Lancashire Regiment (T.F.) 1916-1920, the last two years in India.

Married Doris May Griffin 3.X.1923. No children.

Assistant to Dr P. A. Buxton on London School of Hygiene and Tropical Medicine expedition to Samoa, 1923-1925, and in London helping to write up the results of this expedition until the end of 1926.

Medical entomologist, Kenya Colony, II.1927 - VII.1929.

Entomologist (medical) and subsequently Senior Entomologist (medical), Uganda Protectorate, VII.1929 - I.1948.

Awarded O.B.E. in VI.1947.

Honorary Associate, British Museum (Natural History), 1948-1973.

Recreations: big-game shooting in Africa, butterfly hunting in England.

Died (after long illness): St Paul's Hospital, Hemel Hempstead, Herts., England, 20.II.1973.

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Biographical notes:

Hubbard, C.A., 1971, Ent. News 81(11): 279-284, fig.

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Meillon, B. de, 1972, Ent. News 82(12): 326.

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Obituaries:

Eichler, W. & Timmermann, G., 1974, Angew. Parasit. 15(4): 237-239, 1 fig.

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A c c o n c i s e r e p o r t

Many of the colleagues who unfortunately could not make the journey to England very graciously sent their regards and best wishes for a successful conference. They have our grateful thanks and we hope to have the pleasure of seeing them during a next meeting.

P a r t i c i p a n t s : The following 58 delegates (excluding associated members of families), representing 15 nationalities, attended the Conference:

Dr M.H.Abreu (Lisboa, Portugal)	Dr B.R.Laurence (London, England)
Mr R.Balter (Nottingham, England)	Miss G.Lindqvist (Lund, Sweden)
Dr J.C.Beaucournu (Rennes, France)	Dr J.Löfqvist (Lund, Sweden)
Prof. A.H.Benton (Fredonia, USA)	Dr M.B.Madon (Los Angeles, USA)
Miss E.van den Broek (Utrecht, Neth.)	Mr D.K.Mardon (Aberdeen, Scotland)
Prof. P.Brinck (Lund, Sweden)	Prof. A.Marshall (Aberdeen, Scotland)
Mrs G.Brinck-Lindroth (Lund, Sweden)	Mr A.R.Mead-Briggs (Guildford, England)
Prof. J.R.Busvine (Maidenhead, Engl.)	Mr R.Mehl (Oslo, Norway)
Dr W.Büttiker (Magden, Switzerland)	Dr B.C.Nelson (Berkeley, USA)
Col. D.C.Cavanaugh (Washington, USA)	Prof. R.L.C.Pilgrim (Christchurch, N.Z.)
Dr T. Clay (London, England)	Dr R.Poppenbeek (Surrey Hills, Austr.)
Dr R.E.Doty (Kensington, USA)	Dr M.Portús (Barcelona, Spain)
Dr J.W.Evans (Sydney, Australia)	Dr P.Price (Urbana, USA)
Prof. R.Galun (Ness Ziona, Israel)	Miss H.A.van Rooy (Utrecht, Netherlands)
Mr R.S.George (Duxford, England)	Dr M.Rothschild (Ashton Wold, England)
Mr J.Goose (Saffron Walden, England)	Dr J.Schlein (Jerusalem, Israel)
Dr N.Gratz (Genève, Switzerland)	Dr R.C.H.Shepherd (Frankston, Australia)
Mr M.T.Greenwood (Loughborough, Engl.)	Mr F.G.A.M.Smit (London, England)
Dr G.E.Haas (Palmer, USA)	Mr I.Szabó (Budapest, Hungary)
Mr M.J.Hodson (Berkhamsted, England)	Mr R.N.Taylor (Berkhamsted, England)
Dr G.P.Holland (Ottawa, Canada)	Mr G.B.Thompson (Cambridge, England)
Dr H.Hoogstraal (Cairo, Egypt)	Dr H.V.Thompson (Guildford, England)
Prof. C.E.Hopla (Norman, USA)	Prof. R.Traub (Bethesda, USA)
Dr G.Hosie (Glasgow, Scotland)	Mr P.Tunney (Peterborough, England)
Mr A.M.Hutson (London, England)	Dr J.W.Vinson (Boston, USA)
Dr S.Ito (Boston, USA)	Mr J.Waage (Ascot, England)
Prof. R.C.King (Evanston, USA)	Prof. Dr P.Wenk (Tübingen, Germany)
Prof. Dr H.E.Krampitz (München, Germ.)	Miss B.Williams (Oxford, England)
Mr H.Launay (Rennes, France)	Dr J.Wilson (Oxford, England)

P r o g r a m m e s : Each day the lectures started between 8.30 and 9.00 a.m. and - with a 20 min. coffee-break - lasted till 12.30 - 13.00 hrs. A strict time-schedule for all papers had to be adhered to.

Tuesday 21 June: Registration in afternoon. In the evening an informal buffet dinner at the house, during which Professor Sir Vincent Wigglesworth, in a witty speech, formally opened the Conference.

Wednesday 22 June: Lectures (chairman: Prof. R. Traub) -

R. TRAUB - Some adaptations in fleas.

P. PRICE - General concepts on the evolutionary biology of parasites and the apparently anomalous patterns in the Siphonaptera.

A. MARSHALL - The function of ctenidia in ectoparasitic insects with particular relation to host specificity.

G. P. HOLLAND - The wonderful one-host flea [poem].

R. TRAUB - Notes on the zoogeography of fleas and their hosts.

R. L. C. PILGRIM - The New Zealand flea fauna.

C. E. HOPLA - The zoogeography and host association of *Pulex* in the New World.

J. R. BUSVINE - Fleas in history: fact and fantasy.

Afternoon: a discussion group met for some hours while others enjoyed an excursion to Peterborough Cathedral, Peakirk duck decoy and Burghley House.

Thursday 23 June: Lectures (chairman: Dr N. G. Gratz) -

N. G. GRATZ - Problems and developments in the control of flea vectors of disease.

H. HOOGSTRAAT - Comparison of flea and tick roles in the epidemiology of human diseases.

D. C. CAVANAUGH & J. E. WILLIAMS - Plague: some ecological interrelationships.

B. C. NELSON - Ecology of sylvatic plague in lava caves at Lava Beds National Monument, California.

R. TRAUB - The ecology of murine typhus.

S. ITO & J. W. VINSON - Rickettsiae in flea tissues.

C. E. HOPLA - Fleas as vectors of tularemia organisms in the taiga and tundra areas of Alaska.

R. C. H. SHEPHERD - The European rabbit flea (*Spilopsyllus cuniculi*) in Australia: its use as a vector of myxomatosis in the semi-arid (Mallee) area of Victoria.

A. R. MEAD-BRIGGS & J. R. VAUGHAN - The importance of the European rabbit flea (*Spilopsyllus cuniculi*) in the evolution of myxomatosis in Britain.

J. GOOSE - The control of fleas with Bendiocarb (Ficam).

Afternoon: discussion group; excursion for others to Woodwalton Fen Nature Reserve and Monks Wood Experimental Station.

Evening: a cornucopian cocktail party on the village green (with Morris dancers, visits to Ashton Roman site and Ashton Mill).

Friday 24 June: Lectures (chairman: Prof. P. Brinck) -

R. GALUN - Specificity of the feeding response in fleas and its biological significance.

P. WENK - A comparison of the piercing mechanisms of the mouthparts in some bloodsucking insects.

H. E. KRAMPITZ - *Leptopsylla segnis*: host preference, sessility and mating behaviour of Bavarian strains reared in captivity.

J. SCHLEIN - The structure of the metathorax.

R. C. KING - Recent advances in our understanding of insect vitellogenesis.

F. G. A. M. SMIT - Missing and floating genitalia in male fleas.

G. E. HAAS - Unusual Alaskan fleas.

Afternoon: discussion group; excursion to Cambridge (Fitzwilliam Museum, Botanical Gardens, University Department of Zoology).

Saturday 25 June: Lectures (chairman: Dr T. Clay) -

I. SZABO - The borders between the ranges of certain subspecies of fleas in Hungary.

J. C. BEAUCOURNU - Les Siphonaptères du lapin de garenne *Oryctolagus cuniculus*.

M. H. ABREU - Special aspects of the natural infestation of the rabbit *Oryctolagus cuniculus* in Portugal.

P. BRINCK - Population size of ectoparasites of rodents and shrews.

A. H. BENTON & J. F. DAY - Seasonal changes in the flea fauna of nests of the southern flying squirrel *Glaucomys volans*.

G. BRINCK-LINDROTH - A comparison between the flea fauna of small mammals in the Pyrenees and Scandinavian mountains.

R. MEHL - Fleas from nests of small rodents in Norway.

G. HOSIE - Observations on the occurrence of *Ceratophyllus gallinae* around new housing estates in the west of Scotland.

Afternoon: discussion group while others watched a polo match.

For the evening Dr Rothschild had organised a truly splendid 'Flea Party', with lavish and most delicious food and wine, to the sounds of a pop-group or, indoors, to more refined singing by the Deller Consort. Even though it was a very cold summer evening (someone remarked: "only in England can one see blue people eating red strawberries"), it was a grandiose finale of a conference which has been acclaimed as highly successful.